

Best Exercises for Knee Arthritis: A Complete UK Physiotherapist

UK physiotherapist-approved exercises for knee arthritis. Step-by-step strengthening, flexibility and aerobic exercises to reduce pain and improve mobility.

By **Living With Arthritis Editorial Team** · Pending clinical review ·
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*Person performing knee strengthening exercises on a yoga mat for arthritis physiotherapy
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Knee arthritis affects millions of people across the UK and is the most common reason for joint replacement surgery. Yet research consistently shows that the right exercise

programme can significantly reduce knee pain, improve function and even delay the need for surgery. This comprehensive guide, based on UK physiotherapy guidelines and NICE recommendations, provides a structured exercise programme for knee arthritis.

Why Exercise Is Essential for Knee Arthritis

NICE guidelines state that exercise should be a **core treatment** for all people with knee osteoarthritis, regardless of age, severity, pain level or disability. The evidence is clear: regular exercise reduces knee pain by 25–50% on average (comparable to paracetamol or NSAIDs), strengthens the muscles that support and stabilise the knee joint, improves joint flexibility and range of motion, helps maintain a healthy weight (reducing knee loading), improves balance and reduces fall risk, slows disease progression by supporting cartilage health, and boosts mood and reduces the emotional impact of chronic pain.

Common Concerns

"Will exercise damage my knees?" is the most frequent question physiotherapists hear. The answer is a reassuring **no**. Appropriate exercise does not accelerate cartilage loss or worsen arthritis. In fact, controlled loading of the joint through exercise helps nourish cartilage and maintain its health. The key is choosing the right exercises and progressing gradually.

Strengthening Exercises



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Building the muscles around your knee — particularly the quadriceps, hamstrings and glutes — is the single most important exercise goal for knee arthritis. Strong muscles absorb impact, stabilise the joint and reduce pain.

1. Straight Leg Raises (Quadriceps)

Lie on your back with one knee bent and foot flat on the floor. Keep the other leg straight. Tighten the thigh muscle of the straight leg and slowly lift it to the height of the bent knee (about 30 cm). Hold for 5 seconds. Slowly lower. Repeat 10 times each leg. Do 2–3 sets.

2. Wall Sits (Quadriceps and Glutes)

Stand with your back flat against a wall, feet shoulder-width apart and about 30 cm from the wall. Slowly slide down until your knees reach a 45-degree angle (do not go past 90 degrees). Hold for 10–30 seconds. Slide back up. Rest. Repeat 5–10 times. Increase hold time as you get stronger.

3. Bridges (Glutes and Hamstrings)

Lie on your back with knees bent and feet flat, hip-width apart. Tighten your stomach muscles and squeeze your glutes. Lift your hips towards the ceiling until your body forms a straight line from shoulders to knees. Hold for 5 seconds. Slowly lower. Repeat 10–15 times. Do 2–3 sets.

4. Step-Ups

Use a low step (10–15 cm to start, such as the bottom stair). Step up with the affected leg, pushing through the heel. Bring the other foot up. Step back down, leading with the unaffected leg. Repeat 10 times each leg. Increase step height gradually as strength improves. Hold a handrail for safety if needed.

5. Sit-to-Stand

Sit on a firm chair with feet flat. Without using your hands (cross arms over chest if able), stand up by pushing through your heels. Slowly sit back down with control — don't plop. Repeat 10 times. Do 2–3 sets. This functional exercise directly improves daily activities.

6. Clamshells (Hip Stabilisers)

Lie on your side with knees bent at 45 degrees, feet together. Keeping feet touching, lift the top knee as far as comfortable without rotating your pelvis. Hold 3 seconds. Lower slowly. Repeat 15 times each side. Add a resistance band around the knees when ready for more challenge.

Flexibility and Range-of-Motion Exercises

1. Knee Flexion Stretch

Sit on the edge of a chair. Slide one foot back under the chair as far as comfortable, bending the knee. Hold for 20–30 seconds. Release. Repeat 3–5 times each knee.

2. Knee Extension Stretch

Sit with your leg resting on another chair or stool, knee slightly bent. Gently straighten the knee as much as possible (a small rolled towel under the ankle helps). Hold for 20–30 seconds. Repeat 3–5 times.

3. Hamstring Stretch

Sit on the edge of a chair. Extend one leg in front with the heel on the floor. Keeping your back straight, lean forward from the hips until you feel a gentle stretch behind the knee and thigh.

Hold 20–30 seconds. Repeat 3 times each leg.

4. Calf Stretch

Stand facing a wall, hands on the wall. Step one foot back, keeping it flat. Bend the front knee and lean forward until you feel a stretch in the back calf. Hold 20–30 seconds. Repeat 3 times each leg.

Aerobic Exercise for Knee Arthritis



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Cardiovascular exercise reduces overall inflammation, supports weight management and improves pain tolerance. Aim for 150 minutes per week of moderate activity.

Best Aerobic Options

- Walking — The most accessible. Start with 10 minutes and build up. Wear supportive, cushioned shoes. Use Nordic walking poles for additional support if helpful
 - Swimming — Ideal for knee arthritis as water supports body weight. Front crawl is generally easier on knees than breaststroke (which involves a "frog kick" that can stress the inner knee). Many UK councils offer free or discounted swimming for people with disabilities
 - Cycling — Excellent for knees as it builds quad strength with minimal impact. Set the saddle high enough that the knee only bends slightly at the bottom of the pedal stroke. Stationary bikes remove balance concerns
 - Water aerobics — Combines cardiovascular and resistance exercise in a joint-friendly environment. Many UK leisure centres offer specific classes for joint conditions
 - Elliptical trainer — Low impact, smooth motion that's easier on knees than running or stair climbing
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Exercise Programme Structure

A balanced weekly programme for knee arthritis:

- Strengthening — 2–3 times per week (e.g., Monday, Wednesday, Friday). 20–30 minutes per session
 - Aerobic exercise — 3–5 times per week. 20–45 minutes per session (build up gradually)
 - Flexibility — Daily. 5–10 minutes, ideally morning and evening
 - Rest days — At least 1–2 per week. Rest does not mean immobility — gentle walking and stretching are fine
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Managing Pain During Exercise

- Mild discomfort is normal — Expect some discomfort when starting. Pain that settles within 24 hours is acceptable
- The 2-hour rule — If pain is significantly worse 2 hours after exercise, you did too much. Reduce intensity or duration next session

- Apply warmth before — A warm pack for 10–15 minutes before exercise helps reduce stiffness
 - Apply cold after — An ice pack wrapped in a towel for 10–15 minutes after exercise can reduce any post-exercise swelling
 - Stop if — Pain is sharp, sudden or severe; a joint becomes hot and significantly swollen; you feel unwell
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Getting Physiotherapy Support

NHS physiotherapy is available through GP referral or self-referral in many areas. A physiotherapist can assess your specific knee arthritis, design a tailored exercise programme, teach correct exercise technique, provide manual therapy, use taping or bracing if appropriate, and monitor progress and adapt your programme. Private physiotherapy is also available — look for chartered physiotherapists registered with the CSP.

Consistent, progressive exercise is one of the most effective treatments for knee arthritis — equal to or better than many medications, without side effects. Start today with even a few minutes of the exercises above, and build gradually. Our virtual assistant can help you track your exercise routine and provide encouragement.

This article is for general information only. Consult a physiotherapist or your GP before starting a new exercise programme for knee arthritis.

Knee Exercises for Osteoarthritis

If you have knee osteoarthritis (OA), the priority is loaded strengthening of the quadriceps and glutes combined with low-impact aerobic work. NICE recommends a structured 12-week exercise programme as a **core treatment** — equal in importance to weight management and education. The straight leg raise, wall sit, sit-to-stand and stationary cycling described above are the most studied movements for knee OA and the safest starting points for most adults. Begin at 2 sessions per week, building to 3–4 over six weeks. For a deeper dive into condition-specific pacing, read our osteoarthritis exercises guide and the broader osteoarthritis treatment overview.

Knee Exercises for Rheumatoid Arthritis

Knee involvement in rheumatoid arthritis (RA) calls for a gentler progression and clear flare rules. During an active flare, swap strengthening for range-of-motion only — heel slides, supported knee bends and ankle pumps. Resume strengthening when the joint is no longer warm or significantly swollen. Aquatic exercise (warm-water hydrotherapy at 33–35 °C) is particularly well tolerated. See our rheumatoid arthritis exercises guide for flare-friendly routines and pacing rules.

Gentle Knee Exercises for Severe Arthritis

If you have severe knee arthritis, are awaiting surgery or struggle to weight-bear, these seated and lying options keep the joint moving without overload: seated knee extensions (slowly straighten the knee from a chair, hold 5 seconds, 10 reps), heel slides (lying on your back, slide the heel towards the buttock), and gentle isometric quad sets (press the back of the knee into a rolled towel for 5 seconds). Two short sessions daily of 10 minutes each are more effective and better tolerated than one long session.

Knee Strengthening Exercises After Knee Replacement

After a total or partial knee replacement, your hospital physiotherapist will issue a protocol — follow it precisely for the first 6 weeks. The universal early goals are: regain full knee extension (a straight knee), achieve 90° of knee flexion within 2–4 weeks, and rebuild quadriceps activation through straight leg raises and quad sets. Long-term, the strengthening programme on this page is the standard maintenance routine for most post-replacement patients beyond 12 weeks. Always clear new exercises with your surgical team or chartered physiotherapist first.

Exercises to Avoid With Knee Arthritis

Not all exercise is equal. With knee arthritis, avoid or strictly limit: deep squats below 90°, running on hard surfaces if you have moderate-to-severe OA, high-impact aerobics and jumping, heavy leg extension machines (which load the knee in a vulnerable position), lunges with the front knee travelling far past the toes, and full-depth leg presses. None of these are universally banned, but they are the most common causes of post-exercise flares in our experience. If in doubt, find a physiotherapist near you for a personalised assessment.

Knee Arthritis Exercise FAQ — People Also Ask

What are the best exercises for arthritic knees?

The best exercises for arthritic knees combine three elements: strengthening (straight leg raises, sit-to-stand, wall sits and step-ups), low-impact aerobic work (stationary cycling, swimming or brisk walking) and daily flexibility (heel slides, hamstring stretches and calf stretches). Done 3–4 times per week, this combination reduces knee pain by 25–50% on average — comparable to the effect of paracetamol or NSAIDs.

How often should I do knee exercises for arthritis?

UK physiotherapy guidance recommends strengthening exercises 2–3 times per week, low-impact aerobic exercise 3–5 times per week (30 minutes per session, which can be split into 3 × 10 minutes) and gentle range-of-motion stretches every day. Allow at least one rest day between strengthening sessions to let muscle recover.

Can exercise cure knee arthritis?

Exercise cannot reverse cartilage loss or cure knee arthritis, but it is the single most effective *non-surgical* treatment. A consistent 12-week programme can reduce pain by 25–50%, improve walking distance, delay the need for joint replacement and improve quality of life. The benefits stop if exercise stops — it is a long-term management tool, not a short course.

What exercises should I avoid with knee arthritis?

Avoid high-impact activities (running on concrete, jumping, plyometrics), deep squats below 90°, heavy seated leg-extension machines, and lunges where the front knee travels well beyond the toes. Replace them with stationary cycling, swimming, sit-to-stand from a chair, and partial squats to a chair height.

Are there low-impact knee exercises for severe arthritis?

Yes. Seated knee extensions, heel slides, isometric quad sets, ankle pumps and warm-water (hydrotherapy) exercise are all suitable for severe knee arthritis. Two short 10-minute sessions per day are usually better tolerated than one longer session, and there is no requirement to weight-bear for the exercises to be effective.

How long does it take for knee exercises to reduce arthritis pain?

Most people notice reduced stiffness within 2–3 weeks of consistent practice. Meaningful pain reduction typically appears at 6–8 weeks, with the full benefit of a structured programme seen at 12 weeks. Stopping for more than 2–3 weeks usually reverses the gains.

Evidence-Based 8-Week Progression Timeline

This progression mirrors the structure I use in clinic with patients with knee osteoarthritis and stable inflammatory arthritis, and aligns with **NICE NG226** and current Cochrane reviews on exercise for knee OA. Timelines flex with age, comorbidities, baseline fitness and arthritis type — but the order of progression should not.

Weeks 1–2: Pain Management Phase

Pendulum swings, quad sets and glute sets only. No external resistance. The goal is restoring pain-free movement, not loading. Aim for 2–3 short sessions a day, 10–15 repetitions each.

Weeks 3–4: Strength Foundation Phase

Add isometric quad holds (10 seconds), mini squats limited to a pain-free 0–45° range, standing hamstring curls and supported bilateral calf raises. Daily, 1–2 sets of 10–15 reps. The aim here is muscular endurance, not maximal strength.

Weeks 5–8: Progressive Strengthening Phase

Bodyweight squats through full available range, resistance band work above and below the knee, controlled 4-inch step-ups, static or walking lunges, and single-leg stance for proprioception. Three to four sessions a week, 2–3 sets of 8–12 reps.

Week 8+: Return to Activity

Layer in activity-specific movement — hill walking, gardening, cycling, golf, racket sports — progressing intensity by adding load, speed or duration one variable at a time. A maintenance programme of 2–3 strength sessions a week protects gains long-term.

The Three Form Mistakes I See Most Often

1. Knee valgus (inward collapse)

The knees drift inward during a squat or lunge — usually a sign of weak glutes and poor proprioception, often made worse by rushing progression. Cue "knees track over the second toe," loop a light resistance band above the knees for tactile feedback, and regress to isometric glute work if the collapse persists. This is the most common fault and the one most associated with patellofemoral pain.

2. Knee hyperextension (locking back)

The knee pushes past neutral into a locked position at rest or at the top of a movement. Often a long-standing habit, sometimes quad dominance, sometimes fear of bending. Cue a soft 5–10° bend, use a mirror, and strengthen the hamstrings. Important in arthritic knees because it markedly increases patellofemoral load.

3. Heel rise (weight shifting forward)

Heels lift during a squat and weight shifts onto the toes. Driven by ankle stiffness, quadriceps weakness or poor motor control. Anchor the heels (a piece of tape across the floor works well), strengthen dorsiflexion, and work on ankle mobility. Without fixing this, the knees and hips compensate and overload.

"Good" vs "Bad" Pain — The 2-Hour Rule

The single most useful rule I give patients is the **2-hour rule**: if pain has not settled back to baseline within two hours of finishing exercise, the session was too intense and the next one should be regressed.

Good (therapeutic) pain — keep going

- Mild discomfort during exercise, 0–4/10, that settles within two hours.
- A working-muscle ache or fatigue, not a sharp joint pain.
- No next-day swelling, warmth or stiffness increase.
- Pain trending downward across weeks as strength improves.

Bad pain — stop and reassess

- Sharp or shooting pain (suggests nerve or structural involvement).
 - Pain lasting more than four hours after exercise.
 - Immediate joint swelling or warmth.
 - Pain at rest or waking you at night.
 - Catching, locking or a true feeling of instability.
 - Pain getting worse week-on-week rather than better.
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Modifications for Common Scenarios

Post-operative knees (ACL repair, meniscectomy, knee replacement)

The first 4–6 weeks are a protected weight-bearing phase. Stay with isometric quads and glutes, avoid flexion beyond 45° unless your surgical protocol says otherwise, and use crutches or a frame for as long as recommended. Add resistance from week 6–8. Always work to the surgeon's protocol first.

Severe morning stiffness (inflammatory pattern)

Exercise 1–2 hours after morning medication, when stiffness has eased. Apply heat first, warm up slowly for 5–10 minutes, and split exercise into two or three shorter sessions across the day rather than one long block.

Fear-avoidance (kinesiophobia)

Use graded exposure rather than a normal progression: very short, very frequent sessions (5–10 minutes twice a day) to rebuild confidence. Pair every session with the message that *pain does not equal damage*. Celebrate small wins explicitly — it matters.

Return to sport

Typically 12–16 weeks from symptom onset before sport-specific work. Add lateral movements, controlled plyometrics (step-downs before jumps), and pivoting drills. Test readiness with single-leg squat and Y-balance, then return to play in graded steps (25%, then 50%, then 75% intensity across weeks).

Red Flags — When to Stop and See a GP

Same-day medical attention

- Acute swelling (knee circumference increasing by more than 1cm within 24 hours).
- Heat or redness over the joint.
- The knee giving way during exercise.
- True locking or catching.
- Severe pain >7/10 unrelieved by rest.
- Numbness, tingling or radiating pain down the leg.

Book a GP appointment within 1–2 days

- Increasing pain at rest, or new night pain disturbing sleep.

- Swelling that has not settled within 24 hours.
- A sudden change in pain pattern, or new bruising.

Discuss with your GP if ongoing

- No measurable progress after six consistent weeks — imaging may be appropriate.
 - Pain worsening despite sensible modification — injection or specialist referral may be considered.
 - Significant loss of function affecting daily life.
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Why Patients Stop — and How to Stay With It

The most common reason patients abandon a knee programme isn't pain from the arthritis itself — it's **delayed onset muscle soreness** (DOMS) on days 2–3 of starting. Eccentric loading causes microtrauma; inflammation peaks at 24–72 hours; the patient concludes the exercise is "damaging" them and stops.

What actually helps: knowing in advance that DOMS is normal and not a sign of damage; keeping moving gently on the sore days (walking, easy stretching, heat); and starting with a lower volume than feels necessary so the first wave is mild. By the third session, the body has adapted and the soreness drops sharply. Patients who are pre-warned about DOMS are far more likely to continue.

How to Measure Progress (Beyond Weight and Distance)

I ask patients to track a few of these weekly. Real improvement shows up here long before it shows up on a scan.

- Pain at a specific task (e.g. pain during 10 squats), 0–10 scale, weekly.
- Walking tolerance — minutes before pain onset, or a timed 6-minute walk.
- Daily-activity ease — stairs, sit-to-stand, getting out of a chair, rated 0–10.
- Reps tolerated before fatigue or symptom onset (e.g. 5 squats in week 1 → 30 in week 8).
- Night pain and sleep quality — often the first thing to improve.

- Medication use — frequency of paracetamol or NSAIDs across the week.
- Confidence / fear of movement — informally, or via the TSK-11.

A typical four-data-point trajectory I'd expect over eight weeks: pain during squats 6 → 5 → 3 → 1; walking 10 → 12 → 18 → 25 minutes; stairs 5/10 → 6/10 → 8/10 → 9/10; night pain 4/10 → 3/10 → 1/10 → 0/10. If you're not seeing movement on at least two of these by week 4, the programme needs adjusting — not abandoning.

Related Knee & Arthritis Guides

- Osteoarthritis exercises — full programme
 - Osteoarthritis treatment options (UK)
 - Rheumatoid arthritis exercises
 - More knee exercises for osteoarthritis
 - Knee exercises for rheumatoid arthritis
 - Anti-inflammatory diet for arthritis
 - Find a knee physiotherapist near you
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Q&A

Question: What are the best exercises for arthritic knees?

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